FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

PART TEST –IV
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	B
2.	B	B	D
3.	A	B	A
4.	A	A	B
5.	B	B	D
6.	B	D	C
7.	B	C	B
8.	D	A	C
9.	C	C	D
10.	B	A	C
11.	C	A	A
12.	B	B	A
13.	B	B	D
14.	D	A	A
15.	B, D	A, B, D	A, B, D
16.	B, C	B, C	A, B, C
17.	A, C, D	B, D	A, B, C
18.	A, B	B, D	B, C
19.	C, D	A, C	A, B
20.	D	A, D	B, C

Physics

PART – I

1. Use $\frac{1}{v}+\frac{1}{u}=\frac{2}{R}$. Image after 1st reflection will act as object for 2nd reflection.

2. In inductor current lags voltage by $\pi/2$

3. $$n(4I_0)=\frac{\int_{y=-\frac{\lambda D}{4d}}^{y=\frac{\lambda D}{4d}} 4I_0\cos^2\frac{\pi dy}{\lambda D}dy}{\int_{-\frac{\lambda D}{4d}}^{\frac{\lambda d}{4d}} dy}$$

4. $$E=\frac{hc}{\lambda}=\frac{GM^2}{L}$$

   $\Rightarrow$ dimensions of $\frac{hc}{G}=\frac{LM^2}{L}=M^2$

5. $\langle P\rangle = V_{RMS}I_{RMS}\cos\phi$

14-16. $iB\ell = 0.5$

$\Rightarrow i = \frac{0.5}{B\ell}=\frac{10-Bv\ell}{2}$

$\eta = \frac{(Bv\ell)i}{(0.5)v}$ = efficiency

17. $Q_2 - 20 = 20 - 50$

$\Rightarrow Q_2 = -10$

$U_2 = \frac{Q_2^2}{2C_2}$

18. Energy conservation

$\Rightarrow \frac{Q_1^2}{2C}+\frac{Q_2^2}{2C}+\frac{1}{2}Li^2$ = constant

19. Use energy conservation

Chemistry

PART – II

SECTION – A

2. In $Ni(CO)_4 = Ni \rightarrow 3d^{10}4s^04p^0$ (sp^3 hybridization)
$[Ni(CN)_4]^{2-} = Ni^{2+} \rightarrow 3d^84s^04p^0$ (dsp^2 hybridization)
CN^- is strong field ligand.

3. $HCOOH \rightleftharpoons H^+ + HCOO^-$

$$K_{HCOOH} = \frac{\left[H^+\right]^2}{\left[HCOOH\right]}$$

$CH_3COOH \rightleftharpoons H^+ + CH_3OO^-$

$$K_{CH_3COOH} = \frac{\left[H^+\right]^2}{\left[CH_3COOH\right]}$$

As $[H^+]$ in both cases are equal:

$$\frac{\left[CH_3COOH\right]}{\left[HCOOH\right]} = \frac{K_{HCOOH}}{K_{CH_3COOH}} \Rightarrow \left[CH_3COOH\right] = \frac{1.8\times10^{-4}}{1.8\times10^{-5}}\times10^{-3} = 10^{-2}M$$

5. $Fe^{3+} \rightarrow 3d^53s^0$
$Mn^{2+} \rightarrow 3d^54s^0$

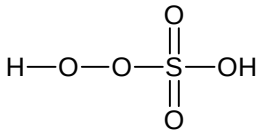
6. $\because PV^{\gamma} = constant$

$$P_0\left(V_0\right)^{\gamma} = P'\left(\frac{V_0}{8}\right)^{5/3}$$

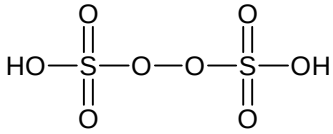
$\Rightarrow P' = 32\ P_0$

16. Irrespective of nature of ligand field, magnetic moment of octahedral complex having metal ions in d^1, d^2, d^8, d^9 configuration, remains same.
Elements may or may not form colourless complex.

17. (B)



(D)

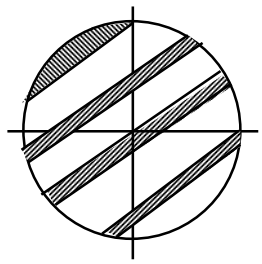


Mathematics PART – III

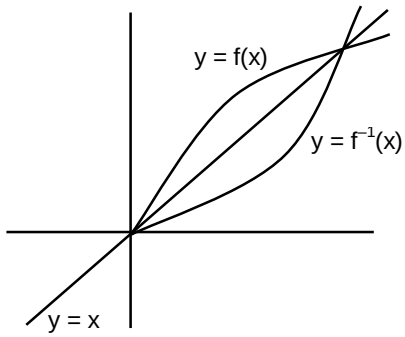
SECTION – A

1. It is clear by symmetry $2A = \pi(3\pi)^2$

$A = \dfrac{9\pi^3}{2}$



2. Clearly A < C < B



3. Circle intersect the $y^2 = 4x$ at $(1, \pm 2)$ end of latus rectum

Required curve = $2\int_0^1 2\sqrt{x} + 2\pi = \dfrac{8}{3} + 2\pi$

4. $\vec{a}.\vec{b} = 0$

5. $|\text{adj}A| = |A|^{n-1}$

$|\text{adj}(\text{adj}A)| = |\text{adj}A|^{n-1} = |A|^{(n-1)^2}$

6. $A^{20} - A^{18} = A\left(A^2 - I\right)$

$A^{18} - A^{16} = A\left(A^2 - I\right)$ ……..

$A^4 - A^2 = A\left(A^2 - I\right)$

$\therefore A^{20} - A^2 = 9A\left(A^2 - I\right)$

$= 9A^3 - 9A = 9A^2 - 9I \Rightarrow \left(\because A\left(A^2 - I\right) = A^2 - I\right)$

$\Rightarrow A^{20} = 10A^2 - 9I$

$\Rightarrow A^{20} = 10\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} - 9I$

$\therefore k + 1 = 10$

$k = 9$

7. Consider S = {1, 2, 3, …..}
Consider event A = selecting the no. 1 from S
Probability of A = $\lim_{n\to\infty} \frac{1}{n} = 0$.
But still. Event A can occur

8. Required vector $= r^3 \times f^3 = \begin{vmatrix} i & j & k \\ -2 & -4 & 1 \\ 3 & 4 & 5 \end{vmatrix}$

10. $M_1^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$, $M_2^k = \begin{bmatrix} x^k & \frac{4(x^k-1)}{x-1} \\ 0 & 1 \end{bmatrix}$
Also, $PP^T = P^TP = I$ and $B = M_1^{2012}, D = M_2^{2012}, E = M_2^{2011}\, M_1^{2011}$

11. Clearly $P(A_1) = \frac{2}{4} = \frac{1}{2}$, $P(A_2) = \frac{2}{4} = \frac{1}{2}$, $P(A_3) = \frac{2}{4} = \frac{1}{2}$,
$P(A_1 \cap A_2) = \frac{1}{4}$, $P(A_2 \cap A_3) = \frac{1}{4}$, $P(A_1 \cap A_3) = \frac{1}{4} = P(A_1 \cap A_2 \cap A_3) = 0$

12. P(A), P(B) are roots of $x^2 - \frac{5}{6}x + \frac{1}{6} = 0$
$\therefore x = \frac{1}{2}, \frac{1}{3}$

P(a,0,0) Q(0,b,0) R(0,0,c)

13. $\therefore \alpha = \frac{a}{3}, \beta = \frac{b}{3}, \gamma = \frac{c}{3}$
$d = \dfrac{\left|\frac{0}{a} + \frac{6}{b} + \frac{b}{c} - 1\right|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$
Locus is $\frac{1}{d^2} = \frac{1}{9\alpha^2} + \frac{1}{9\beta^2} + \frac{1}{9\gamma^2}$
$\therefore$ Locus of α, β, γ is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{d^2}$

14. Equation of plane is $lx + my + nz = p$ (say)
$A\left(\frac{p}{l}, 0, 0\right), B\left(0, \frac{p}{m}, 0\right), C\left(0, 0, \frac{p}{n}\right)$
Let (α, β, γ) is a point in locus
$\therefore \alpha = \frac{p}{\ell}, \beta = \frac{p}{m}, \gamma = \frac{p}{n}$
$\therefore \ell^2 + m^2 + n^2 = 1$

15. $D = 3abc - (a^3 + b^3 + c^3)$
$D_1 = D_2 = D_3 = -(a^2 + b^2 + c^2 - ab - bc - ac)$
As plane are distinct then $a \neq b \neq c$
Point of intersection $\left(\frac{1}{a+b+c}, \frac{1}{a+b+c}, \frac{1}{a+b+c}\right)$

The lines $\frac{x-1}{a}=\frac{y}{b}=\frac{z+1}{c}$ and $\frac{x+1}{c}=\frac{y}{b}=\frac{z-1}{a}$ have shortest distance zero.

The lines $\frac{x-1}{a}=\frac{y-1}{b}=\frac{z-1}{c}$ and $\frac{x+1}{b}=\frac{y+1}{c}=\frac{z+1}{c}$ will intersect if a = b = c which is not possible

16. (A) $\left(P^T\right)^T = P$, P^T is orthogonal

$\Rightarrow \left(P^T\right)^T P^T = I$

$\Rightarrow PP^T = I$

$\Rightarrow$ P is orthogonal

(B) P is orthogonal

$\Rightarrow PP^T = I$

$\Rightarrow P^{-1}PP^T = P^{-1}I$

$\Rightarrow P^T = P^{-1}$

$\Rightarrow \because P^T$ is orthogonal $\Rightarrow P^{-1}$ is orthogonal

(C) $P^T = P^{-1}$ and $Q^T = Q^{-1}$

$\therefore (PQ)(PQ)^T = PQQ^TP^T = PQQ^{-1}P^{-1} = PIP^{-1} = PP^{-1} = I$

$\therefore (PQ)^T = (PQ)^{-1}$

So, PQ is orthogonal

17. $A^2 = -I$, $A^3 = -A$, $A^4 = I$ also $I + A + A^2 + A^3 = 0$

18. $\Delta = \sin(A-B)\sin(B-C)\sin(C-A) = 0$

$\Rightarrow$ Either A = B or B = C or C = A or A = B = C

Hence, the result follows

20. $y - f(x) = f'(x)(y - x)$ and $y - g(x) = g'(x)(y - x)$

$\Rightarrow \frac{f'(x)}{f(x)} = \frac{g'(x)}{g(x)} \Rightarrow f(x) = \frac{1}{4}g(x) \Rightarrow f(x) = e^{x/4}, g(x) = 4e^{x/4}$

$I = \int_{-1}^{1} \frac{x^2+4}{4+4e^{x/4}}dx = \frac{13}{12}$

Area $\int_0^1 (4e^{x/4} - e^{x/4})dx = 12(e^{1/4} - 1)$